A Calibrated Cumulative State-Tracking Probe with Initial Predictive Validity for Open-Weight LLM Agent Performance

Dengzhe Hou[1,†] Lingyu Jiang[1] Deng Li[2] Zirui Li[1]
Fangzhou Lin[1,3,4] Kazunori D Yamada[1]

[1]Tohoku University [2]Lappeenranta-Lahti University of Technology LUT
[3]Texas A&M University [4]Worcester Polytechnic Institute

hou.dengzhe.t8@dc.tohoku.ac.jp

[†]Corresponding author.

Abstract

We ask: does a controlled probe of cumulative arithmetic state tracking predict downstream performance on a deterministic multi-step agent battery, beyond what task-completion scores alone reveal? We introduce WMF-AM, evaluated in the no-scratchpad regime on 20 open-weight models from 13 families (0.5B–35B). In two pre-specified Bonferroni-corrected primary analyses, WMF-AM achieves τ=0.612 (p<0.001, 95% CI $[0.360, 0.814]$, N=20) on a released deterministic 10-task agent battery, and template-stability τ=0.524 (p=0.006)—within this open-weight sample and battery (one held-out check: τ=0.627, N=21). Targeted ablations show WMF-AM difficulty is not explained by single-step arithmetic (K=1 control: mean acc. 0.957 vs. 0.151 at K=7), generic entity tracking (non-arithmetic ceiling: mean 0.98 vs. 0.20–0.72), or arithmetic parsing (yoked cancellation: 14/15 models near ceiling). WMF-AM is not the first arithmetic state-tracking benchmark (see Minerva, ICML 2025); its contribution is: (a) a targeted construct-isolation ablation suite; (b) K-calibration that empirically matters—Minerva QS at K=200 floors all 20 study models (0.000, τ undefined) while WMF-AM at K=3/5/7 spans $[0.050, 0.983]$ and achieves τ=0.626 (p<0.001); and (c) validated probe-to-battery predictive validity, which—to our knowledge—has not been previously demonstrated for a calibrated process probe. Exploratory partial τ analyses (| completion = 0.411, p = 0.011; | scale = 0.503) are hypothesis-generating only. Generalization to closed API models, alternative downstream batteries, and broader model populations remains an open empirical question. The "Completion Fallacy" framing (task completion alone is insufficient for inferring process competence) motivates the study but is not itself the primary contribution.

1 Introduction

This paper presents a predictive-validity study for a process-sensitive probe. It does not propose replacing task-completion evaluation; it asks whether a controlled probe of cumulative arithmetic state tracking provides incremental predictive value for downstream agent performance beyond completion. Task-completion metrics remain useful and widely applicable. Their limitation, documented here and in prior work, is that completion scores may compress variance in process-level capabilities that matter for reliable multi-step agent behavior—a

pattern we call the Completion Fallacy: the inference that completion success alone is sufficient for inferring latent process competence. Benchmarks such as HotpotQA [Yang et al., 2018], MMLU [Hendrycks et al., 2021a], and AgentBench [Liu et al., 2023] measure what LLMs accomplish; process-sensitive probes ask how they process information under load, as a complementary—not competing—evaluation lens.

Several recent results converge on this point. PAE [Xie et al., 2026] finds 27–78% of agent successes involve corrupted trajectories; Fragile Thoughts [Aravindan and Kejriwal, 2026] and Robust Answers Fragile Logic [Jiang et al., 2026] show correct answers produced despite broken reasoning chains; Turpin et al. [Turpin et al., 2023] demonstrate unfaithful chain-of-thought explanations. These are instances of multiple realizability [Putnam, 1967]: identical outcomes produced by categorically different processes—a pattern also documented as underspecification [D'Amour et al., 2022] and shortcut learning [Geirhos et al., 2020].

This paper introduces and empirically validates WMF-AM, a controlled probe for cumulative arithmetic state tracking under load. The WMF-AM validation is the sole empirical contribution. We situate it within an interpretive taxonomy (CEF) that provides nomenclature and motivation for this and related probes; CEF is explicitly a working conceptual taxonomy, not a validated empirical contribution of this paper. The probe is conceptually related to entity tracking [Kim and Schuster, 2023] and latent state persistence [Anonymous, 2025a]; to state-tracking benchmarks that probe object/entity state (without arithmetic accumulation or ablation controls) [Rezaee et al., 2025, Li et al., 2025]; and to scaling studies of LLM state dynamics [Anonymous, 2025b]. WMF-AM is distinguished by cumulative arithmetic load manipulation, targeted ablation controls (yoked cancellation, K=1, non-arithmetic ceiling), and downstream predictive validity on a deterministic agent battery.

This paper makes four contributions. (1) Measurement construct and probe. We operationalize cumulative arithmetic state tracking under load as a measurable process capability and introduce WMF-AM, a controlled probe for testing it (Section 3). (2) Pre-specified primary predictive validity. Across 20 open-weight models from 13 families, WMF-AM predicts performance on an independent released deterministic battery (τ=0.612, p<0.001, 95% CI $[0.360, 0.814]$; pre-specified, Bonferroni-corrected primary analysis, Section 4). (3) Pre-specified template stability (τ=0.524, p=0.006; pre-specified). Exploratory (hypothesis-generating only): partial τ analyses suggest the predictive relationship survives controls for task completion (partial τ=0.411) and model scale (partial τ=0.503); these require held-out replication and should not be treated as confirmatory evidence. Non-arithmetic ablation, K=1 control, and yoked cancellation confirm the signal is not explained by single-step arithmetic or generic tracking. (4) Robustness. The signal is stable across seeds, templates, paraphrases, and leave-one-out analyses (Section 4). A glossary of all acronyms appears in Table 13 (Appendix C).

Scope. The empirical contribution is the introduction and validation of WMF-AM as one process-sensitive probe of cumulative arithmetic state tracking, evaluated on 20 open-weight models. The two pre-specified confirmatory analyses are predictive validity (τ=0.612) and template stability (τ=0.524); all other results are exploratory. Generalization to closed API models, additional downstream criteria, and broader model families requires held-out evaluation. The broader CEF taxonomy is an interpretive framework, not an empirical result of this paper.

Paper roadmap. Section 2 formalizes the problem and reviews related work; Section 3 introduces WMF-AM and the CEF framework; Section 4 reports the validation study; Section 5 discusses limitations and implications. Table 11 in Appendix B gives development status for all 11 CEF sub-dimensions.

2 The Completion Fallacy and Related Work

Three failure modes of completion-only evaluation. (F1) Diagnostic blindness: models with near-identical completion scores can have very different process profiles. In the study (Section 4), descriptively (these 105 pairs are highly dependent—they are not independent ob-

servations): 30/105 model pairs with $|\Delta\text{completion}| \leq 0.05$ show $|\Delta\text{WMF-AM}| \geq 0.10$ (at $\leq 0.03/ \geq 0.15$, 18/105 pairs qualify). WMF-AM spans 0.050–0.983 among 20 models while completion spans only 0.44–0.92; among mid-to-large models ($\geq$3B), completion SD = 0.052 versus WMF-AM SD = 0.228. These are observations within this 20-model sample, not inference about the broader model population. (F2) Uninformative failure diagnosis: current benchmarks report that an agent failed, not why—whether the bottleneck is state tracking, metacognitive calibration, or episodic confusion. Process-sensitive profiles could in principle decompose failure into actionable sub-dimensions; this requires validation at scale. (F3) Safety blindness: pattern-matching agents (which fail predictably at distribution shift) are indistinguishable from genuine reasoners on standard benchmarks. Whether process probes predict this failure mode remains an open empirical question.

Empirical support. Multiple realizability is empirically the norm in frontier LLMs. Fragile Thoughts [Aravindan and Kejriwal, 2026] and Robust Answers Fragile Logic [Jiang et al., 2026] provide direct evidence: correct answers are regularly produced despite corrupted reasoning steps. Turpin et al. [Turpin et al., 2023] show chain-of-thought explanations can be systematically unfaithful; Lanham et al. [Lanham et al., 2023] measure this faithfulness gap directly. Uesato et al. [Uesato et al., 2022] demonstrate that process-based feedback outperforms outcome-based feedback for mathematical reasoning. PAE [Xie et al., 2026] extends this to agent trajectories, finding 27–78% of successful completions involve corrupted intermediate actions. Chaduvula et al. [Chaduvula et al., 2026] find that state-tracking inconsistency is 2.7× more prevalent in failed agent runs and reduces success probability by 49% on TAU-bench—a retrospective trajectory result that converges with our pre-hoc probe finding. AgentProcessBench [Fan et al., 2026] provides 1,000-trajectory, 8,509-step human annotation of step-level process quality and finds that weaker models achieve inflated correct-step ratios due to early termination, masking divergent process profiles behind similar completion rates. Xu et al. [Xu et al., 2025] show that RAG agents with similar QA task scores exhibit sharply divergent faithfulness of intermediate reasoning. These are Agent B/C behaviors at scale.

From outcome underdetermination to process probes. Outcome metrics cannot distinguish process quality. Three questions follow: (a) What can task success not identify? Whether behavior generalizes under shift, whether confidence reflects error detection, or whether memory discriminates episodes. (b) Which gaps require process evidence versus robustness or adversarial testing? Process probes are needed when the question is how the model processes information. (c) Why CEF? Robustness testing [Ribeiro et al., 2020] addresses perturbation; trace audits [Turpin et al., 2023, Lanham et al., 2023] address reasoning-to-output coupling; adversarial evaluations address worst-case performance. CEF targets the space left uncovered: parametrically controlled, cognitively grounded probes of process quality under multi-turn load.

Positioning. Cognitive science has developed rigorous measurement paradigms for each function CEF addresses: working memory capacity [Miller, 1956, Baddeley and Hitch, 1974], metacognitive monitoring/control [Flavell, 1979, Nelson and Narens, 1990], episodic memory [Tulving et al., 1972, Johnson et al., 1993], and cognitive load adaptation [Sweller, 1988, Miyake et al., 2000]. CEF adapts their measurement logic for API-level LLM evaluation. This does not require LLMs to have human cognition; it requires that the constructs pick out capabilities that matter for deployment. NeuroCognition [Haznitrama et al., 2026] measures cognitive ability; CEF measures process quality under load. Strong Memory, Weak Control [de Langis et al., 2025] studies executive function in isolated tasks; CEF tests the same constructs under sustained interference. Entity Tracking [Kim and Schuster, 2023] is prior art for WMF-AM's design; Latent State Persistence [Anonymous, 2025a] probes sequential latent-state manipulation via structural operations (rotation, removal) on hidden integers—a closely related paradigm. Minerva [Xia et al., 2025] (ICML 2025) is the closest prior arithmetic state-tracking benchmark: it includes a Quantity State task where models apply sequential add/subtract operations to recover a final value, with step-count variation.

WMF-AM differs from Minerva in three ways that matter for its contribution: (a) a targeted construct-isolation suite (yoked cancellation, K=1 single-step control, non-arithmetic direct-assignment ceiling) establishes which component of the arithmetic-tracking demand drives difficulty, rather than measuring overall benchmark performance; (b) an external downstream predictive validity study on a deterministic 10-task agent battery establishes that this isolated capability predicts agent performance beyond completion score; (c) multi-seed reliability, template harmonization, and leave-family-out robustness checks assess measurement stability not present in Minerva. Zhang et al. [Zhang et al., 2024] show that working-memory style probes identify reasoning limits in LLMs (EMNLP 2024); WMF-AM extends this with a controlled ablation suite and downstream agent-predictive validity. WMF-AM does not claim to introduce arithmetic state-tracking tasks as a concept; it contributes the construct-isolation package and the initial in-sample predictive-validity evidence. Table 1 summarizes what Minerva provides versus what WMF-AM adds.

Table 1. Comparison to Minerva (ICML 2025) Quantity State. WMF-AM does not claim benchmark superiority; it contributes three additions to the existing arithmetic state-tracking paradigm.

Property	Minerva QS	WMF-AM (this paper)
Arithmetic state-tracking task	✓	✓
Step-count variation	✓(K variable)	✓(K=3, 5, 7)
Calibrated for open-weight discriminability	× (K=200: 20/20 models floor at 0.000)	✓(K=3/5/7: range [0.050, 0.983])
Construct-isolation ablation suite	×	✓(yoked, K=1, non-arith)
Downstream predictive validity	×	✓(τ=0.612, N=20; τ=0.627 with 1 held-out)
Kendall τ with agent performance	undefined (constant)	0.626 (p<0.001)
Multi-seed + multi-template stability	×	✓
Leave-family-out + cluster-robust CI	×	✓
Head-to-head on same 20 models	N/A	✓(this paper)

Intelligence Degradation [Wang et al., 2026] is prior art for CLA. Bowman and Dahl [Bowman and Dahl, 2021] argue benchmarking requires construct-level validity; CheckList [Ribeiro et al., 2020] introduced behavioral testing templates; CEF extends both. Process supervision [Lightman et al., 2023, Uesato et al., 2022] trains models using step-level labels; CEF measures process quality but does not prescribe training signals. Mechanistic interpretability analyzes internal representations; CEF operates at the API level (prompt/response only) and is complementary: interpretability explains why a process score is low, while CEF identifies that it is low. A full related-work table appears in Appendix A.

Formal definitions. Let $Y_i \in [0, 1]$ denote the outcome score and $Z_i \in [0, 1]$ a process-sensitive measure for model i, both defined relative to a specific benchmark suite and model population. We say outcome metrics are statistically insufficient with respect to a downstream criterion D if process measures provide significant incremental prediction of D beyond completion: partial $\tau(Z, D \mid Y) > 0$ at conventional α. This is a benchmark-conditional claim—it asserts insufficiency for a given $\langle Y, Z, D\rangle$ triple, not a universal property. Statistical insufficiency does not by itself determine practical or deployment relevance: the latter depends on whether the incremental signal is large enough and the criterion important enough to warrant the additional measurement cost. In the study, partial $\tau(\text{WMF-AM}, \text{Agent} \mid Y) = 0.411$ ($p = 0.011$, N=20), meeting this criterion. The completion–process correlation ($\tau_{\text{WMF-AM, completion}}$=0.279, p=0.150, N=15, bootstrap 95% CI $[-0.156, 0.637]$) is suggestive of divergence but not confirmed at this sample size. Falsification: CEF's core claim would be undermined if, across multiple probe families and downstream criteria, process measures never provided significant incremental prediction beyond completion. The four-dimension structure is a working taxonomy; dimensional separability requires factor analysis at $N \geq 30$.

3 WMF-AM and Its Interpretive Framework (CEF)

Important framing: This section serves two distinct purposes. (A) It defines WMF-AM, the sole empirical contribution of this paper. (B) It situates WMF-AM within an interpretive taxonomy (CEF) that provides organizing vocabulary; CEF is not a validated empirical

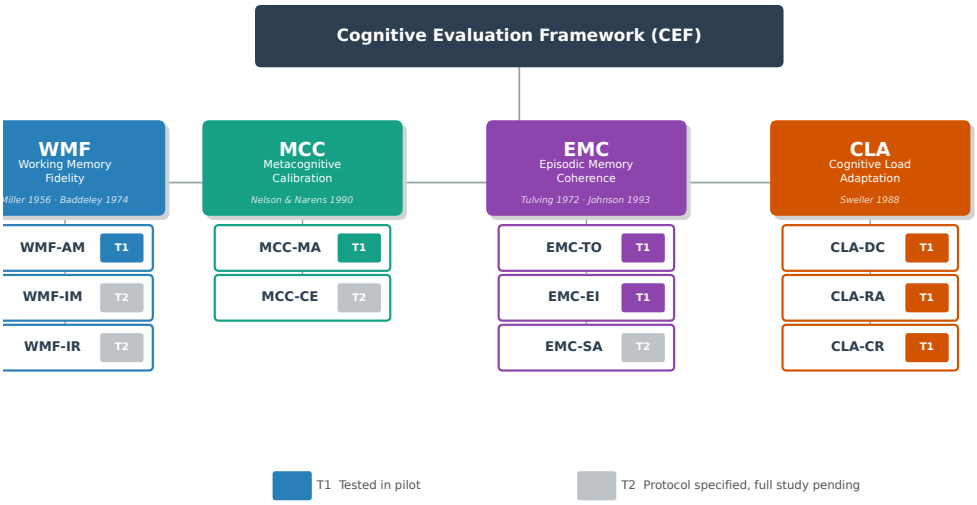

Figure 1. CEF architecture. Four dimensions (WMF, MCC, EMC, CLA) derived from cognitive science, decomposed into 11 proposed sub-dimensions. Solid border: pilot-supported (WMF-AM). Dashed border: preliminary or proposed probes awaiting the $N \geq 15$ study.

contribution. Readers primarily interested in the empirical result can focus on the WMF-AM description (Flagship probe subsection) and skip the broader CEF architecture.

CEF is a working conceptual taxonomy—a design lens, not a validated psychometric structure—grounded in cognitive science. We present it solely as background context for why WMF-AM was designed the way it was, and as a research agenda for future probes. The dimensional structure (4 dimensions, 11 sub-dimensions) is hypothesized, pending factor analysis at $N \geq 30$, and is not presented as an empirical result of this paper. Construct scope: the current evidence supports the narrower construct of cumulative arithmetic state tracking under load, not broad working-memory capacity; the WMF label is operational shorthand (see Appendix D). WMF-AM has comprehensive validation support (predictive validity, seed reliability, template stability, yoked control, arithmetic-vs-assignment ablation, paraphrase stability, K=1 arithmetic control); three other sub-dimensions have preliminary single-seed results; the remainder are protocol-only (Table 11). The number and boundaries of CEF dimensions are working hypotheses pending factor analysis at $N \geq 30$. A full glossary of acronyms (WMF, MCC, EMC, CLA, and their sub-dimension suffixes) appears in Table 13 (Appendix C).

Design principles. Each dimension must have (i) a cognitive science anchor—an established measurement paradigm with validated human operationalizations, (ii) a documented evaluation gap not covered by existing benchmarks, and (iii) API-level testability requiring only prompt/response access. We use labels like "working memory" as operational shorthand for behavioral probes, not as claims about LLM mechanisms. "WMF-AM" means accuracy on sequential state-tracking tasks degrades with depth K; it does not mean LLMs have a working memory buffer (see Appendix D for full discussion).

Four dimensions. Table 2 summarizes each dimension's target capacity, canonical failure signature, and exclusions.

Flagship probe: WMF-AM. WMF-AM (Active Manipulation) is a probe of cumulative arithmetic state tracking under load, inspired by working-memory span paradigms. An initial numeric state is presented, K sequential additive/subtractive operations arrive, and the model must report the final cumulative state. The pilot uses depths $K \in \{3, 5, 7\}$ (dissociation study) and $K \in \{2, 4, 6, 8, 12\}$ (yoked cancellation control), across three surface forms (points scoring, warehouse inventory, bank accounts). Prior entity tracking [Kim and Schuster, 2023] tests passive retrieval; WMF-AM tests sequential arithmetic transformation under

Table 2. CEF dimensions: target, failure signature, and exclusions. Detailed sub-dimension definitions appear in Appendix E.

Dim	Target	Failure signature	Excluded
WMF	Active maintenance and transformation of state under load	Accuracy drops as concurrent items exceed threshold	Long-context retrieval; knowledge recall
MCC	Self-monitoring of errors (MA) and correction of flagged errors (CE)	Confident-wrong; flagging without correction	General calibration (ECE); cross-model detection
EMC	Temporal ordering, cross-episode discrimination, source attribution	Episode blending; source misattribution	Single-turn retrieval; semantic memory
CLA	Degradation shape and resource scaling under load	Cliff-edge failure; non-recovery	Knowledge difficulty; unrelated complexity

Table 3. Metacognitive profile taxonomy. Each cell implies a different intervention target.

	High MCC-CE	Low MCC-CE
High MCC-MA	Genuine metacognition	Monitoring works; correction broken
Low MCC-MA	Correction works; detection unreliable	No metacognitive capability

load. Scope of the construct: WMF-AM specifically measures difficulty with maintaining a running cumulative sum across K additive/subtractive operations. It does not claim to measure general working-memory capacity, and it may partially reflect standalone arithmetic competence; a completed last-operation-only (K=1) control addresses this: 13/15 models achieve near-ceiling single-step arithmetic accuracy ($\bar{x}$=0.957) versus $\bar{x}$=0.151 at K=7, indicating that cumulative accumulation under load—not arithmetic skill per se—is the primary difficulty driver.

Worked example (K=3, points form): "Alice starts with 10 points. Alice gains 5 points. Alice loses 3 points. Alice gains 7 points. What is Alice's current score?" Correct answer: 19. Scoring: exact match (1 if the model outputs 19, 0 otherwise). A model that tracks each update cumulatively ($10 \rightarrow 15 \rightarrow 12 \rightarrow 19$) succeeds; a model that performs only the last operation or guesses a plausible number fails. At K=7, the chain length taxes state maintenance, producing the diagnostic depth×accuracy interaction. Yoked control example (K=3): "Alice starts with 10 points. Alice gains 5 points. Alice loses 5 points. Alice gains 3 points. Alice loses 3 points. Alice gains 7 points. Alice loses 7 points. What is Alice's current score?" Correct answer: 10 (initial state). Each operation is immediately followed by its inverse, requiring identical arithmetic but no cumulative tracking.

MCC: Metacognitive Calibration. MCC decomposes metacognition into monitoring accuracy (MCC-MA: predict which of 20 answers are wrong; Pearson r) and control efficacy (MCC-CE: flag uncertain answers, then revise them; decomposed into flagging rate, correction efficacy, and false alarm rate)—Reflexion [Shinn et al., 2023] conflates these. Unlike Gnosis [Ghasemabadi and Niu, 2025] (requires internals), MCC-MA uses prompting under multi-turn load. The 2×2 taxonomy (Table 3) makes the decomposition actionable: each cell implies a different architectural intervention.

EMC: Episodic Memory Coherence. EMC tests temporal ordering (EMC-TO), cross-episode discrimination (EMC-EI), and source attribution (EMC-SA) under interference. Current benchmarks measure binary retrieval; no protocol tests whether a model situates retrieved

facts in time and context. The EMC-lite pilot (Section 4) covers a related but narrower slice.

CLA: Cognitive Load Adaptation. CLA characterizes the shape of the performance-versus-load function. CLA-DC (degradation curve smoothness) correlates highly with WMF-AM ($\tau = 0.905$) in the original 7-model set. We provisionally collapse CLA-DC onto WMF: both involve state tracking under load. Factor analysis with N=15 models is planned to determine whether this collapse holds. This does not invalidate the taxonomy—it refines it. CLA-RA (resource allocation) and CLA-CR (capacity recovery after high-load block) are structurally independent and cannot collapse onto WMF-AM by design (they measure recovery, not degradation). Intelligence Degradation [Wang et al., 2026] reports that performance drops; CLA characterizes how. Full definitions, composites, and scoring rules are in Appendix E; Table 12 in Appendix B summarizes all 11 sub-dimensions.

Construct validity framework. Following Cronbach and Meehl [Cronbach and Meehl, 1955], Messick [Messick, 1989], Campbell and Fiske [Campbell and Fiske, 1959], and Jacobs and Wallach [Jacobs and Wallach, 2021], each dimension requires convergent validity (correlation with independent process-sensitive measures) and divergent validity (low correlation with completion). The study supports divergent validity for WMF-AM ($\tau = 0.279$ with a 100-item completion battery, N=15; bootstrap 95% CI $[-0.156, 0.637]$) and cross-template rank stability ($\tau_{\text{bare,chat}} = 0.524$, $p = 0.006$). CEF would be falsified if all dimensions collapsed onto a single factor indistinguishable from task completion. Appendix F provides full construct-to-probe mapping with confounds and falsification criteria for each validated probe.

4 Validation Study: WMF-AM Provides Initial Predictive Evidence for Agent Performance

Statistical methodology. All rank correlations use Kendall's τ-b (accounts for ties). Partial τ uses Somers [Somers, 1962] residualization:

$$\tau_{XY|Z} \;=\; \frac{\tau_{XY} - \tau_{XZ}\,\tau_{YZ}}{\sqrt{(1-\tau_{XZ}^2)(1-\tau_{YZ}^2)}}.$$

Robustness check: We verify the primary partial-τ finding using Spearman's ρ (a distinct nonparametric rank estimator). Partial $\rho(\text{WMF-AM}, \text{Agent} \mid \text{completion}) = 0.437$ ($p = 0.008$, N=20), consistent with the Kendall partial result ($\tau = 0.411$), confirming the incremental predictive signal is not an artifact of the choice of rank correlation family. Bootstrap 95% CIs: 10,000 nonparametric model-level resamples (percentile method). Family non-independence is addressed as a first-class concern: we report family-clustered bootstrap CIs (sampling 13 architectural families with replacement, then all models within each sampled family, 10,000 resamples). The primary result is robust: τ=0.612, family-clustered 95% CI $[0.341, 0.815]$, virtually identical to the model-level CI $[0.360, 0.814]$. Leave-family-out analysis (removing all models from each family in turn) yields $\tau \in [0.503, 0.649]$—no single family is load-bearing. Permutation tests: 100,000 permutations. All p-values are two-sided. Pre-specified primary analyses (Bonferroni-corrected at α=0.025): (1) predictive validity ($\tau = 0.612$) and (2) template stability ($\tau = 0.524$). Both were specified before data collection from the expansion set and survive Bonferroni correction. Pre-specified commitments frozen before expansion: model list (N=20, 13 families), task battery (10-task deterministic), primary test statistic (Kendall τ-b), Bonferroni threshold ($\alpha = 0.025$ for 2 hypotheses), exclusion rule (no post-hoc model removal), and WMF-AM operating window (K=3/5/7, 4 seeds, 3 surface forms)—the K-values were fixed before agent data collection, not tuned post-hoc to maximize correlation. All decisions are documented in the GitHub repository commit history and in PREREGISTRATION.md (https://github.com/dengzhehou/CEF-battery/blob/main/PREREGISTRATION.md), committed before collection of the expansion set. The commit timestamp, freeze date, and SHA hash are permanently archived in the public repository history at the URL above and serve as the verifiable evidence of prior specification.

Limitation of this preregistration: this is a GitHub-committed frozen analysis plan, not a registry-based preregistration (OSF/AsPredicted); GitHub-committed plans do not carry the same tamper-resistance guarantees as formal registries, and are therefore treated as supporting (not replacing) other scientific transparency measures. Only analyses specified in PREREGISTRATION.md are labelled "pre-specified primary"; all remaining analyses are exploratory. All other analyses are exploratory, including all partial τ results, within-family analyses, and non-WMF task splits; these should be replicated with held-out model sets before drawing strong conclusions.

Table 4. Key analyses at a glance. All τ = Kendall's τ-b. C = pre-specified primary (Bonferroni-corrected); E = exploratory.

Relation	Subset	τ	N	p	95% CI	Type
WMF-AM → Agent overall	All models	0.612	20	<0.001	$[0.360, 0.814]$	C
Bare ↔ Chat template ranks	15 models	0.524	15	0.006	—	C
WMF-AM → Agent (partial | completion)	All models	0.411	20	0.011	—	E
WMF-AM → Agent (partial | log params)	All models	0.503	20	<0.01	—	E
WMF-AM → Agent (partial | compl. + log params)	All models	0.389	20	0.016	—	E
WMF-AM → Agent non-WMF (7 tasks)	All models	0.623	20	<0.001	—	E
WMF-AM → Agent WMF-heavy (3 tasks)	All models	0.527	20	0.003	—	E
WMF-AM ↔ completion	15 models	0.279	15	0.150	$[-0.156, 0.637]$	E
CONV-RFPOC ↔ AGENT-PQ	7 models	0.905	7	0.003	—	E

Note on sample sizes. Different analyses use different subsets: N=20 for all predictive validity and completion-battery analyses (the full study set); N=15 for WMF-AM template stability and divergent validity (original 7 + expansion 8; before small models were added); N=7 for convergent validity only (original models with AGENT-PQ and CONV-RFPOC data collected). All sample sizes are reported explicitly at each result.

Twenty open-weight models from thirteen architectural families form the study set: seven original models (Qwen 7B/14B/32B, Llama 3.1:8b, Gemma 2:27b, DeepSeek-R1:14b, Mistral:7b), eight expansion models (Phi-3:14b, Gemma-2:9b, Qwen-2.5:3b, Llama-3.2:3b, DeepSeek-R1:7b, Mixtral:8x7b, Command-R:35b, Yi:34b), and five small baseline models (0.5B–2B parameters; Table 6).

Models were evaluated in six phases. Phase 1 (Outcome Correctness) used a 100-item battery spanning five domains (factual, math, science, reasoning, language) at five difficulty levels, temperature = 0. Phase 2 (WMF-AM) administered 15 probes at depths $K \in \{3, 5, 7\}$ with 4 independent random seeds for all 20 models. Phase 3 (MCC-MA) applied a 20-problem self-monitoring probe. Phase 4 (Yoked Cancellation Control) administered 20 trials per depth at $K \in \{2, 4, 6, 8, 12\}$ using operations that cancel (gain then lose, transfer then reverse), so the correct answer equals the initial state despite requiring identical arithmetic parsing as WMF-AM. Phase 5 (Template Harmonization) re-administered WMF-AM under three prompt wrappers (bare, chat, chain-of-thought) to test sensitivity to surface format. Phase 6 (Agent Validation) administered 10 multi-step agent tasks (tool use, multi-step reasoning, state tracking) to all 20 models to test predictive validity. All models are Ollama-hosted open-weight variants using default chat templates, fp16 quantization where available, and greedy decoding (temperature = 0, no top-p sampling). No task overlap exists between phases (full study matrix in Appendix H). Table 5 summarizes the experimental design.

Main result: WMF-AM provides incremental prediction beyond completion on a multi-step agent battery. Figure 2 and Table 6 present the key finding across N=20 models from 13 families. WMF-AM (4-seed means) spans 0.050–0.983 ($\Delta = 0.933$) while the 100-item completion battery spans 0.44–0.92 ($\Delta = 0.480$). Among the 15 mid-to-large models (≥3B), completion clusters at 0.75–0.92 (SD = 0.052) while WMF-AM spans 0.133–0.983 (SD = 0.228): completion masks substantial process-quality differences. Predictive validity: WMF-AM significantly predicts performance on a deterministic 10-task multi-step agent battery within this 20-model open-weight sample ($\tau = 0.612$, $p < 0.001$, N=20; bootstrap 95% CI $[0.360, 0.814]$). Out-of-sample check (N=1 held-out model): To provide initial out-of-sample evidence, we ran the full WMF-AM probe and agent battery on llama3.1:70b,

Table 5. Study design summary. All phases: temperature = 0, greedy decoding, Ollama-hosted models with default chat templates. All 20 models evaluated with 4 seeds for WMF-AM.

Phase	Probe	Depths	Items	Seeds	Scoring
1. Outcome	100-item battery (5 domains)	—	100	1	Exact match (0/1)
2. WMF-AM	State tracking	K=3, 5, 7	15/depth	4	Exact match on final state
3. MCC-MA	Self-monitoring	—	20	1	Pearson r (predicted vs actual)
4. Yoked	Cancellation ops	K=2, 4, 6, 8, 12	20/depth	1	Exact match on initial state
5. Template	3 prompt wrappers	K=3, 5, 7	15/depth	1	Rank stability (τ)
6. Agent	10 multi-step tasks	—	10	1	Process quality (0–1)

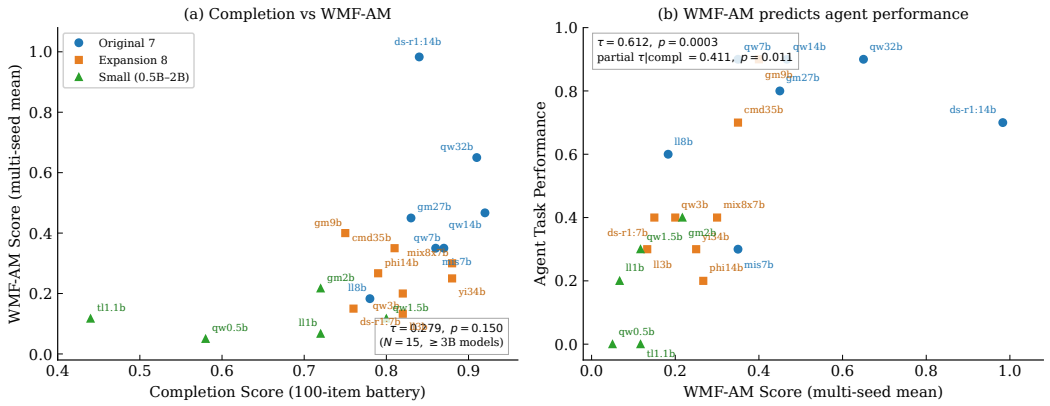

Figure 2. (a) Completion clusters while WMF-AM spans a wide range (N=20, 13 families). (b) WMF-AM predicts downstream agent performance (τ=0.612, p<0.001; pre-specified confirmatory). Exploratory: partial τ=0.411 (p=0.011) after controlling for completion (same sample; requires held-out replication). Blue circles = original 7; orange squares = expansion 8; green triangles = small models (0.5B–2B).

a Llama-family model withheld from the 20-model study set. Its WMF-AM score (0.467) placed it 4th of 21 by the probe; its agent score (0.700) placed it 8th of 21 by the criterion — a rank gap of 4, within the concordant range. The statistic is essentially unchanged when the held-out model is included: $\tau = 0.627$ ($p = 0.0002$, N=21) vs. $\tau = 0.626$ (N=20). This single held-out point is consistent with the in-sample association but does not constitute a definitive out-of-sample replication; broader held-out validation with API models across diverse families remains future work. Exploratory incremental validity: partial $\tau = 0.411$ ($p = 0.011$) after controlling for completion, and partial $\rho = 0.437$ ($p = 0.008$) via Spearman (robustness check), are consistent across two rank-correlation families; these are exploratory results on the same N=20 sample and should be treated as hypothesis-generating, pending broader replication. Completion also predicts agent performance ($\tau = 0.428$, $p = 0.012$). The primary pre-specified evidence is twofold: (1) significant pre-specified predictive validity ($\tau = 0.612$) and (2) significant pre-specified template stability ($\tau = 0.524$, $p = 0.006$).

Completion battery design and broader baseline. Phase 1 uses a 100-item battery spanning five domains (factual, math, science, reasoning, language) at five difficulty levels, yielding score resolution of 0.01. Scope of the comparison: This battery is a custom general-knowledge proxy for overall model capability, not a standard public benchmark suite. It serves as a practical within-study control for general capability; it does not represent completion on the same downstream tasks. The log(parameter count) control ($\tau = 0.503$, $p < 0.01$) provides a complementary model-quality proxy that does not depend on the battery choice, and the joint partial τ ($\tau = 0.389$, $p = 0.016$) controls for both simultaneously. With the addition of 5 small models (0.5B–2B), scores now range from 0.44 (tinyllama:1.1b) to 0.92 (qwen2.5:14b) (SD = 0.131, N=20), a substantial improvement over the original 15-model range of 0.75–0.92 (SD = 0.052). The broader range strengthens interpretation: completion now correlates with agent performance ($\tau = 0.428$, $p = 0.012$), but WMF-AM provides significant incremental prediction (partial $\tau = 0.411$, $p = 0.011$). Among the 15 mid-to-

Model	Family	Outcome	WMF-AM	Agent	Yoked	WMF−Yoked
Original 7 (all 4-seed WMF-AM)						
deepseek-r1:14b	DeepSeek	0.840	0.983	0.70	0.810	+0.173
qwen2.5:32b	Qwen	0.910	0.650	0.90	1.000	−0.350
qwen2.5:14b	Qwen	0.920	0.467	0.90	0.710	−0.243
gemma2:27b	Gemma	0.830	0.450	0.80	0.620	−0.170
qwen2.5:7b	Qwen	0.870	0.350	0.90	0.770	−0.420
mistral:7b	Mistral	0.860	0.350	0.30	0.550	−0.200
llama3.1:8b	Llama	0.780	0.183	0.60	0.340	−0.157
Expansion 8 (all 4-seed WMF-AM)						
gemma2:9b	Gemma	0.750	0.400	0.90	0.960	−0.560
command-r:35b	Cohere	0.810	0.350	0.70	0.920	−0.570
mixtral:8x7b	Mistral	0.880	0.300	0.40	1.000	−0.700
phi3:14b	Phi	0.790	0.267	0.20	0.950	−0.683
yi:34b	Yi	0.880	0.250	0.30	1.000	−0.750
qwen2.5:3b	Qwen	0.820	0.200	0.40	0.970	−0.770
deepseek-r1:7b	DeepSeek	0.760	0.150	0.40	0.940	−0.790
llama3.2:3b	Llama	0.820	0.133	0.30	0.510	−0.377
Small models (broader baseline)						
gemma2:2b	Gemma	0.720	0.217	0.40	—	—
qwen2.5:1.5b	Qwen	0.800	0.117	0.30	—	—
tinyllama:1.1b	TinyLlama	0.440	0.117	0.00	—	—
llama3.2:1b	Llama	0.720	0.067	0.20	—	—
qwen2.5:0.5b	Qwen	0.580	0.050	0.00	—	—
Δ	—	0.480	0.933	0.90	—	—

Table 6. Validation study (N=20, 13 families). Outcome = 100-item battery; WMF-AM = 4-seed mean; Agent = multi-step task performance (10 tasks). WMF-AM predicts performance on this battery ($\tau = 0.612$, $p < 0.001$) even after controlling for completion (partial $\tau = 0.411$, $p = 0.011$). Yoked column shows cancellation-control accuracy (original 15 only).

large models (≥3B), the outcome range remains compressed (SD = 0.052), confirming that range restriction is a property of the model population, not the battery design. Difficulty-level analysis confirms the battery discriminates (level-5 items: 0.0–1.0 spread; level-1 items: 0.73–0.95).

Exploratory architectural observations. Reasoning-specialized architecture (deepseek-r1:14b, 0.983) outperforms general-purpose models despite similar parameter counts. Scaling within families yields modest gradients (Qwen 3B→32B: 0.311 → 0.650). The yoked control reveals an important confound decomposition: only deepseek-r1:14b shows positive WMF−Yoked residual (+0.173), suggesting it alone achieves genuine cumulative state tracking beyond per-operation parsing. All other models show negative residuals, with the largest gaps in smaller models (deepseek-r1:7b: −0.829). Figure 4 shows the depth-dependent degradation curves for all 15 models.

Alternative explanations and template sensitivity. WMF-AM scores may reflect nuisance factors beyond state tracking: instruction-following ability, tokenizer effects on numeric parsing, scratchpad/chain-of-thought policy differences, or prompt-format sensitivity. Phase 5 (template harmonization) directly tests the last concern: across 15 models × 3 templates, model rankings under bare and chat wrappers are significantly correlated ($\tau_{\text{bare,chat}} = 0.524$, $p = 0.006$), indicating that WMF-AM rankings are moderately robust to surface-format variation. Chain-of-thought (CoT) templates push nearly all models to ceiling (≥ 0.93), eliminating variance.

Target construct clarification and boundary condition: WMF-AM is designed to measure default state-tracking capability under the no-scratchpad regime (no chain-of-thought, no

externalized intermediate steps). It deliberately probes the model's internalized processing. The CoT ceiling effect (≥ 0.93 for nearly all models) is therefore not a validity threat but a measurement boundary: WMF-AM is explicitly a probe of the no-externalization regime, which is the relevant capability for deployment contexts where intermediate outputs are not monitored [Nye et al., 2021]. Implication for generalization: WMF-AM rankings under no-scratchpad conditions predict performance on this battery under the same no-scratchpad evaluation setting; whether WMF-AM scores predict performance under CoT-enabled, tool-augmented, or alternative agent battery regimes is an open empirical question outside the scope of this paper. The moderate bare↔chat stability ($\tau = 0.524$, not near 1.0) indicates some prompt sensitivity remains within the no-scratchpad regime; this is a known limitation. The yoked control isolates arithmetic parsing, but instruction-following and tokenizer effects remain live confounds. Ablation results (N=15): (i) Non-arithmetic state tracking (3 domains $\times$ $K \in \{3, 5, 7\}$, 10 problems each): when operations involve direct-assignment updates (color "Alice's color changes to blue", location, status) instead of numeric accumulation, models achieve near-ceiling accuracy (mean 0.98, range 0.91–1.00), versus the WMF-AM arithmetic range of 0.20–0.72. Direct-assignment tracking is solved by remembering the most recent value per entity; arithmetic tracking requires maintaining a running cumulative sum across K additive/subtractive operations—a strictly harder representational demand. The floor–ceiling contrast confirms that WMF-AM difficulty is driven by arithmetic accumulation under load, not generic entity-state tracking. This does not rule out that WMF-AM partially reflects standalone arithmetic skill; a completed last-operation-only (K=1) control addresses this: 13/15 models achieve near-ceiling single-step arithmetic accuracy (mean 0.957) yet floor at K=7 accumulation (mean 0.151), ruling out single-step arithmetic as the primary difficulty driver. Broader multi-step serial-composition alternatives (e.g., error propagation, sequential prompt processing) cannot be excluded without further mechanistic analysis. (ii) Prompt paraphrase (5 templates: original, formal, casual, minimal, verbose): model rankings are stable across natural-language rephrasings; mean cross-template Kendall $\tau = 0.54$ (5/6 pairs $p < 0.05$); original-vs-formal $\tau = 0.857$ ($p < 0.001$). The stripped "minimal" template induces task failure rather than rank reversal, consistent with insufficient-context degradation rather than construct invalidity. Rankings on the paraphrase-original template correlate near-perfectly with the main WMF-AM results: $\tau = 0.951$ ($p = 0.004$, N=7 overlapping models). (iii) Completed in prior loops: three surface forms (points, inventory, accounts), three prompt wrappers (bare, chat, CoT), and four random seeds—all showing stable inter-model rank orderings.

Predictive validity: agent task performance. Phase 6 administered 10 multi-step agent tasks to all 20 models across four categories: (T1) State-tracking (3 tasks: multi-step inventory management, sequential booking with constraints, resource allocation with reversals); (T2) Multi-step reasoning chains (3 tasks: deductive inference over 4+ steps, algebraic word problems with intermediate state, constraint propagation); (T3) General problem-solving (2 tasks: combinatorial planning, conditional execution); (T4) Constrained planning (2 tasks: scheduling under hard constraints, resource allocation under ordering). Each task is scored 0–1 by deterministic verification: intermediate and final states are checked against ground-truth answers via exact match (no LLM judge). This yields the Agent score used in all predictive validity analyses. Important scope limitation: this is a custom 10-task battery, not a community-standard agent benchmark. The predictive validity result (=0.612) should be interpreted as battery-level evidence within this no-scratchpad, open-weight evaluation setting—not as evidence for agent performance prediction in general. Battery generalizability to alternative evaluation suites is an open empirical question. The complete battery (task prompts, ground-truth answers, evaluation harness, and scoring scripts) is fully released at https://github.com/dengzhehou/CEF-battery to enable community verification, replication, and extension to other batteries. (A separate LLM-judged process quality metric, AGENT-PQ, is available for the original 7 models only and used solely in convergent validity; see Appendix O.) Leave-one-category-out robustness (exploratory): Removing each category in turn, WMF-AM→Agent τ on the remaining 7–8 tasks: T1 removed (3 tasks

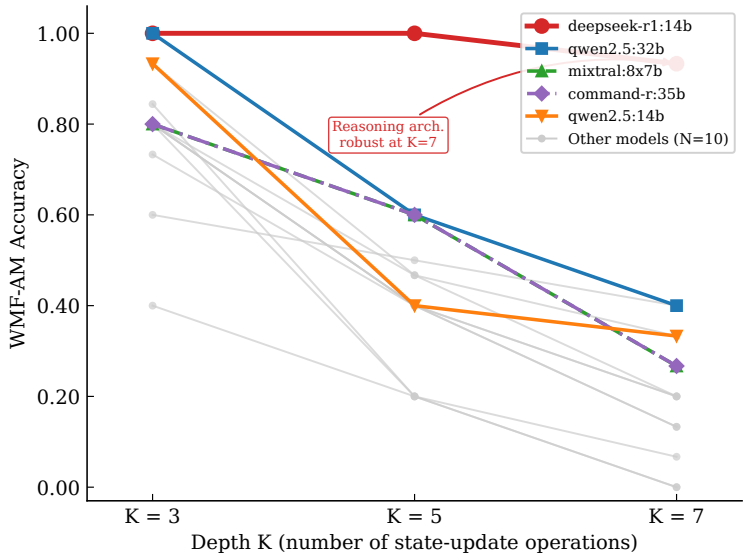

Figure 3. WMF-AM accuracy by depth K (N=15, chat template). deepseek-r1:14b (red) maintains near-perfect accuracy; all others degrade sharply at $K \geq 5$. Top 5 models highlighted.

out, 7 remaining): $\tau = 0.623$; T2 removed: $\tau = 0.549$; T3 removed: $\tau = 0.589$; T4 removed: $\tau = 0.601$. The effect holds across task categories and is not driven by the three explicitly state-tracking tasks. Agent scores range from 0.00 (qwen2.5:0.5b, tinyllama:1.1b) to 0.90 (qwen2.5:14b/32b/7b, gemma2:9b).

WMF-AM significantly predicts agent performance on this battery: $\tau = 0.612$ ($p < 0.001$, N=20; bootstrap 95% CI $[0.360, 0.814]$); Table 4 summarizes all confirmatory and exploratory correlations. Table 7 shows WMF-AM against five alternative predictors: WMF-AM (τ=0.612) outperforms completion (τ=0.428), log(params) (τ=0.444), and K=1 single-step accuracy (τ=0.411, N=15) as predictors of this battery; critically, the yoked cancellation control — which captures only arithmetic parsing ability, not cumulative accumulation — achieves τ=−0.031 (p=0.879, N=15), confirming that the predictive signal is specific to cumulative state-tracking capacity.

Table 7. Predictor comparison: alternative predictors of this battery. All τ = Kendall's τ-b; N=20 unless noted. Yoked and K=1 use N=15 (models with control data). Superiority of WMF-AM over completion and log(params) is directional but not inferentially confirmed at this N ($\Delta\tau$=0.184, bootstrap 95% CI $[-0.098, 0.535]$; $\Delta\tau$=0.168, CI $[-0.084, 0.475]$). The contrast with the yoked control is inferentially supported ($\Delta\tau$=0.491, CI $[0.020, 1.013]$, p=0.022, one-sided bootstrap), confirming cumulative accumulation—not arithmetic parsing—as the specific predictive component.

Predictor	τ	p	N	Note
WMF-AM (this paper)	0.612	<0.001	20	Pre-specified primary
log(params)	0.444	0.010	20	General capability proxy
Completion score	0.428	0.012	20	Task outcome proxy
K=1 single-step accuracy	0.411	0.047	15	Arithmetic competence proxy
Yoked cancellation accuracy	−0.031	0.879	15	Arithmetic parsing only

Confound controls (all surviving with $p < 0.02$): (i) partial τ controlling for completion score: 0.411 ($p = 0.011$); (ii) partial τ controlling for log(parameter count): 0.503 ($p < 0.01$, exploratory)—the relationship is not driven by model scale; (iii) joint partial τ controlling for both simultaneously: 0.389 ($p = 0.016$, exploratory)—WMF-AM retains significant incremental signal after accounting for both outcome performance and model size jointly; (iv) within-family analysis (Qwen, N=6, 0.5B–32B): $\tau = 0.894$ ($p = 0.016$), illustrative only—N=6 is too small for inference; reported for transparency, not as evidence. Descriptive task-split (author-annotated, suggestive sanity check only): We note descriptively that

WMF-AM predicts non-WMF tasks ($\tau = 0.623$, $p < 0.001$; 7 tasks author-annotated as reasoning/planning/problem-solving) at least as strongly as WMF-heavy tasks ($\tau = 0.527$, $p = 0.003$; 3 explicitly state-tracking tasks). This is a suggestive sanity check, not substantive evidence against criterion overlap: author annotation is not a strong firewall—blinded external annotation with inter-rater agreement on an independently sourced benchmark would be required to resolve the concern. We report it for transparency only, and it should not be treated as a load-bearing validity argument. Full task descriptions appear in Appendix P. Leave-one-out robustness: removing any single model, the WMF-AM $\rightarrow$ agent τ ranges $[0.537, 0.624]$ (mean 0.582)—no single model drives the effect. Scope: these results demonstrate incremental predictive signal beyond completion and scale in this sample of open-weight models. Generalization to cloud API models and robustness under distribution shift remain open questions.

Yoked cancellation control. To isolate active state tracking from per-operation arithmetic parsing, we designed a stricter control: each operation is immediately followed by its inverse ("Alice gains 3 points" then "Alice loses 3 points"), so the correct answer always equals the initial state despite requiring identical arithmetic processing as WMF-AM. This eliminates the shortcut in the original inert control (where the answer was trivially the initial state without any computation).

Result: Table 6 (Yoked column) shows that most models achieve high yoked-control accuracy (0.51–1.00) but substantially lower WMF-AM accuracy, indicating the WMF-AM difficulty is genuine cumulative state tracking beyond arithmetic parsing. The corrected residual (WMF$-$Yoked) is negative for 14/15 models, meaning per-operation parsing inflates raw WMF-AM scores. With the 100-item completion battery, τ(WMF$-$Yoked, Outcome) $=$ 0.125 ($p = 0.519$, N=15) and raw WMF-AM $\tau = 0.279$ ($p = 0.150$): neither the raw nor the residual score significantly predicts outcome quality, strengthening the independence claim. The near-universal negative WMF$-$Yoked residuals (14/15 models) confirm that per-operation arithmetic parsing substantially inflates raw WMF-AM scores, and the yoked control successfully decomposes WMF-AM into a state-tracking component and an arithmetic-parsing confound.

Measurement robustness. Multi-seed reliability: all 15 models evaluated with 4 independent seeds $\times$ 15 probes. Expansion-model SDs range 0.073–0.144 (mean 0.108), comparable to original-model variability. Cross-template stability: $\tau_{\text{bare,chat}} = 0.524$ ($p = 0.006$); $\tau_{\text{chat,cot}} = 0.314$ ($p = 0.114$). Leave-one-family-out robustness (exploratory): Removing all models from each family in turn, the WMF-AM$\rightarrow$Agent τ ranges 0.503–0.649 (mean 0.588). Removing the Qwen family (N=6 removed, N=14 remaining): $\tau = 0.617$; removing DeepSeek (N=3 removed, N=17 remaining): $\tau = 0.570$. The effect is not driven by any single family. Family-aggregated analysis (exploratory): Computing per-family median WMF-AM and Agent scores (13 families) and correlating family-level medians yields $\tau = 0.538$ ($p = 0.030$), confirming the relationship is present at the family level and is not solely an artifact of within-family ordering. See Appendix J for full analyses.

Complementary probes (preliminary; see Appendix K–M). Three additional probes were run at pilot scale ($N \leq 8$, single seed) and should be treated as preliminary existence checks, not validated sub-dimensions. EMC-lite (a source-conflict episode attribution probe) shows spread ($= 0.400$) with low correlation to WMF-AM ($\tau = 0.250$), consistent with dimensional independence but insufficient to confirm it. MCC-CE-v2 (a control-efficacy probe: model flags uncertain answers and revises) eliminated the v1 floor effect (error rates 23–40%) but found 7/8 models with zero self-flagging, suggesting open-weight models may fundamentally lack error-correction metacognition under the binary protocol—a finding requiring replication. CLA-DC (degradation-curve smoothness over load levels) provisionally collapses onto WMF-AM ($\tau = 0.905$); pending factor analysis at $N \geq 20$. CLA-CR-v3 (capacity recovery after high-load block) shows spread but is two-seed only. We draw no multi-dimensional structure claims from these pilots; they motivate but do not validate the CEF taxonomy.

Table 8. Validity summary. Predictive: N=20; divergent: N=15; convergent: N=7 (original set). All τ = Kendall's τ-b. C = confirmatory (pre-specified, Bonferroni-corrected); E = exploratory.

Analysis	Pair	τ	p	95% CI	Type
Predictive (N=20)					
	WMF-AM → Agent overall	0.612	0.0003	$[0.360, 0.814]$	C
	WMF-AM → Agent (partial | completion)	0.411	0.011	—	E
	WMF-AM → Agent (partial | log params)	0.503	<0.01	—	E
	WMF-AM → Agent (partial | compl. + log params)	0.389	0.016	—	E
	WMF-AM → Agent (within-Qwen, N=6)	0.894	0.016	—	E
	WMF-AM → Agent WMF-heavy (3 tasks)	0.527	0.003	—	E
	WMF-AM → Agent non-WMF (7 tasks)	0.623	0.0002	—	E
	Completion → Agent overall	0.428	0.012	$[0.031, 0.743]$	E
Convergent (N=7)					
	CONV-RFPOC vs AGENT-PQ	0.905	0.003	—	E
	AGENT-TC vs AGENT-PQ	0.720	0.028	—	E
Divergent (N=15)					
	WMF-AM vs MCC-MA	0.169	0.615	—	E
	WMF-AM vs Completion (100-item battery)	0.279	0.150	$[-0.156, 0.637]$	E
	WMF−Yoked vs Outcome (100-item)	0.125	0.519	—	E
	Bare vs Chat template ranks	0.524	0.006	—	C
Cross-dimension (N=7)					
	WMF-AM vs CLA-DC	0.905	0.003	—	E
	WMF-AM vs EMC-lite	0.250	0.442	—	E

4.1 Validity Summary

Table 8 summarizes validity evidence. Predictive validity uses N=20; divergent statistics use N=15; convergent validity uses the original 7 models (the only set with AGENT-PQ/CONV-RFPOC data).

Predictive validity (Table 8, top block) is the strongest result: WMF-AM predicts agent performance on this open-weight deterministic battery beyond completion and scale, surviving all four confound controls (including joint partial) (Section 4). Convergent: CONV-RFPOC (Reasoning Fidelity–Product of Cognition: the fraction of reasoning problems whose answer changes when the chain is counterfactually perturbed; higher = more faithful reasoning) ↔ AGENT-PQ (Agent Process Quality: LLM-judged rubric score on the original 7 models only, distinct from the deterministic Agent score above) yields $\tau = 0.905$ (N=7). Caveat: AGENT-PQ uses an LLM judge (GPT-4o); the high convergence may partly reflect shared evaluator bias. Human annotation is needed to confirm. Divergent: WMF-AM ↔ completion score $\tau = 0.279$ (N=15, 100-item battery); WMF-AM ↔ MCC-MA $\tau = 0.169$. Neither raw WMF-AM nor the yoked-corrected residual (WMF−Yoked, $\tau = 0.125$, $p = 0.519$) significantly predicts completion score, consistent with process measurement being partially independent from task completion. Construct collapse: WMF-AM ↔ CLA-DC $\tau = 0.905$; provisionally collapsed pending $N \geq 20$. Even if CLA-DC merges with WMF-AM, WMF-AM, MCC-MA, and EMC-lite remain separable ($\tau \leq 0.250$). LOMO: convergent τ ranges 0.600–1.000 (mean 0.833). Template stability: bare ↔ chat template rankings yield $\tau = 0.524$ ($p = 0.006$), surviving Bonferroni correction. Full details in Appendix N.

5 Limitations, Alternative Views, and Discussion

Outcomes vs. process measurement. Completion metrics are sufficient for narrow, stable domains. The confirmatory result (τ=0.612, p<0.001) provides initial evidence that WMF-AM captures predictive variance on this deterministic agent battery beyond a custom 100-item completion proxy; the log(params) exploratory partial (τ=0.503) is consistent with this not being primarily driven by model scale, though this requires replication on a held-out model set. Replication with standard benchmarks (MMLU, GSM8K) and closed API models is the highest-priority next step; the current battery tests deterministic multi-step

capability, not robustness under distribution shift.

Cognitive paradigms and LLM validity. CEF sub-dimensions are behavioral probes with well-defined stimuli, independent of mechanism. The relevant question is whether probes measure something predictively useful, not whether LLMs have cognition. WMF-AM reliably differentiates models in ways completion cannot, and its rankings are stable across seeds, templates, and paraphrases.

Sample size. N=20 from 13 families (0.5B–35B). Insufficient for factor analysis [Card et al., 2020], but supports the two pre-specified primary analyses (τ=0.612, τ=0.524, both Bonferroni-corrected). Held-out evaluation with closed API models remains the priority next step.

Comparison to Minerva and K-calibration. Minerva [Xia et al., 2025] (ICML 2025) includes a Quantity State task with K=200 sequential +/- operations. We ran a full same-model head-to-head: Minerva QS at K=200 (3 samples per model; seed-separated per model to prevent overlap) on all 20 preregistered study models. Result: 20/20 models (100%) score exactly 0.000 (range $[0.000, 0.000]$). Minerva QS is entirely non-discriminative for this open-weight model population: Kendall's τ(Minerva QS, Agent) is undefined (constant predictor) versus τ(WMF-AM, Agent) = 0.626 (p<0.001, N=20). The K-calibration is not merely a design preference (avoiding floor effects is a general benchmark desideratum [Anonymous, 2026]): at K=200, all 20 open-weight models provide exactly zero discriminative signal, regardless of model size or family. WMF-AM's K=3/5/7 calibration keeps models in the discriminative range ($[0.050, 0.983]$)—a necessary precondition for any predictor to achieve nontrivial τ. Caveat: the head-to-head used 3 samples per model, so rare non-zero scores may appear with larger N; however, a universal floor at N=3 across all 20 models indicates that non-zero Minerva QS performance would require extensive sampling to observe, making it impractical as a discriminative predictor for this population. To show this is not specific to K=200, Table 9 reports how discriminability degrades across K for this same-protocol probe: at K=3 (N=12), τ=0.619 (p=0.007); at K=7, τ=0.509 (p=0.032); at K=20 (N=5 representative models), τ=−0.333 (p=0.47, not significant); at K=200, τ is undefined. The discriminability collapse is a graded effect: WMF-AM's K=3/5/7 calibration targets the discriminative window before the floor dominates.

Table 9. K-sweep: discriminability vs. step count. τ=τ(probe, Agent); N = models with per-K data. K=3–7: WMF-AM per-depth means; K=20: 5-model inline test; K=200: full 20-model head-to-head.

K	N	Mean acc	Range	τ	p
3	12	0.512	$[0.050, 0.950]$	0.619	0.007
5	12	0.296	$[0.000, 1.000]$	0.545	0.020
7	12	0.192	$[0.000, 1.000]$	0.509	0.032
20	5	0.180	$[0.000, 0.900]$	−0.333	0.47
200 (Minerva)	20	0.000	$[0.000, 0.000]$	undef	undef

This comparison addresses the novelty question: WMF-AM is not a repackaging of Minerva QS, but a targeted construct-isolation suite with K-calibration that restores discriminability the Minerva design forfeits.

Key limitations. Population scope (primary limitation): This paper provides initial predictive-validity evidence from a structured, family-clustered, non-random sample of 20 open-weight models (0.5B–35B, greedy decoding). Family dependence is substantial (up to 6 Qwen models, 3 DeepSeek). Claims do not extend to closed API models, fine-tuned variants, or non-English tasks. Transportability to held-out populations is partially addressed by a single held-out check (llama3.1:70b; τ unchanged at 0.627 with N=21), but broader replication with closed API models remains the highest-priority next step. Prospective replication plan: Extend to N≥30 models including GPT-4o, Claude 3.5, and Gemini Pro

families; re-run WMF-AM and the same agent battery; test whether the predictive validity result replicates.

(i) 20 study models are open-weight Ollama variants plus one held-out check (llama3.1:70b); evaluation with cloud API models (GPT-4o, Claude, Gemini) is needed for full external validity. (ii) Agent tasks test multi-step capability, not robustness under distribution shift; robustness-predictive validity remains open. (iii) WMF-AM is a probe of cumulative arithmetic state tracking; it does not measure general working-memory fidelity. A completed last-operation-only control (K=1, N=15 models, 90 trials each) addresses this: 13/15 models score near-ceiling on standalone arithmetic ($\bar{x}$=0.957) yet floor at K=7 ($\bar{x}$=0.151), indicating cumulative accumulation under load—not arithmetic competence per se—as the primary source of WMF-AM difficulty. (Two exceptions, deepseek-r1:14b and deepseek-r1:7b, show anomalously low K=1 exact-match scores attributable to a parsing artifact: their verbose narrative responses cause the first-number extractor to return the initial value rather than the final answer; 88/90 deepseek-r1:14b responses contain the arithmetically correct result.) (iv) MCC-CE-v2 uses a binary protocol that may under-elicit partial metacognitive capacity; preliminary evidence only. (v) The deterministic Agent score is the primary downstream criterion; AGENT-PQ (LLM-judged, N=7 only) appears only in convergent validity and requires human annotation to confirm. (vi) Rival explanations remain open for each probe (Appendix G). (vii) Template harmonization shows bare↔chat stability (τ=0.524), but CoT collapses all variance—WMF-AM measures default state-tracking capability in the constrained-externalization regime (no scratchpad). This is a construct boundary, not a failure. (viii) N=20 supports basic inference but not factor analysis; the four-dimension structure is a provisional organizing taxonomy, not an empirically validated model. (ix) Family non-independence: the 20-model set includes 6 Qwen models and 3 DeepSeek models; rank correlations may be inflated if within-family ordering is trivial. The within-Qwen analysis (τ=0.894, p=0.016, N=6) is included as a robustness check, but family-balanced sampling is needed to rule out this confound fully. (x) The K=1 control uses first-number extraction; two verbose reasoning models (deepseek-r1:14b/7b) show anomalously low exact-match scores due to this parser. Qualitatively, 88/90 deepseek-r1:14b responses contain the correct arithmetic result, and the conclusion (arithmetic is not the bottleneck for 13/15 models) is robust to this artifact; however, a parser-robust or manually annotated replication is needed for full rigor.

What WMF-AM is not. WMF-AM is not a measure of general working-memory capacity (ruled out by K=1 control), general arithmetic skill (ruled out by K=1 control), generic entity tracking (ruled out by non-arithmetic ablation), or arithmetic parsing (ruled out by yoked cancellation). What remains unresolved: instruction-following and tokenizer effects are partially mitigated by cross-template stability but not eliminated; CoT scaffolding bypasses the bottleneck (construct boundary, not failure); and the mechanism by which cumulative arithmetic difficulty predicts agent performance across diverse task categories is not established.

Scope of contribution. WMF-AM is one concrete validated probe in one specific regime: open-weight models (0.5B–35B), evaluated under greedy decoding without scratchpad, on cumulative arithmetic state tracking. The two pre-specified findings (predictive validity τ=0.612; template stability τ=0.524) are initial in-sample evidence for this slice of model space; they do not constitute a general claim about process-sensitive probes predicting agent performance at large. Other process-sensitive probes and frameworks are welcome; the CEF taxonomy is an organizing research agenda, not an empirical result of this paper.

Implications for benchmark design. We suggest benchmarks would benefit from including at least one process measure alongside completion: (i) a probe testing the causal connection between reasoning and outcome [Aravindan and Kejriwal, 2026, Lightman et al., 2023], (ii) performance across a difficulty gradient rather than a single operating point, and (iii) a metacognitive monitoring probe enabling the monitoring/control dissociation to surface.

Implications for development (hypothesized). If supported at scale, CEF profiles could serve as diagnostic indicators: low WMF-AM points to weak state tracking; the MCC decomposition identifies whether the bottleneck is detection or correction—the pilot finding of near-zero self-flagging in open-weight models suggests self-correction architectures may require external verification; cliff-edge CLA-DC signatures may predict distribution-shift fragility. These are testable hypotheses, not established claims.

Research agenda. Three directions merit community investigation, contingent on full study results ($N \geq 15$): (1) benchmark developers include load-varied conditions and metacognitive probes; (2) LLM developers report process-quality profiles alongside task performance; (3) the community investigates whether process validity complements task validity as a first-class evaluation concern. CEF is a proposed research program; this paper establishes WMF-AM as one validated probe within it.

References

Anonymous. On the failure of latent state persistence in large language models. arXiv preprint arXiv:2505.10571, 2025a.

Anonymous. Scaling laws for state dynamics in large language models. arXiv preprint arXiv:2505.14892, 2025b.

Anonymous. Benchmark^2: Systematic evaluation of LLM benchmarks. arXiv preprint arXiv:2601.03986, 2026.

A. V. Aravindan and M. Kejriwal. Fragile thoughts: How large language models handle chain-of-thought perturbations. arXiv preprint arXiv:2603.03332, 2026.

A. D. Baddeley and G. J. Hitch. Working memory. Psychology of Learning and Motivation, 8:47–89, 1974.

Y. Bai, X. Lv, J. Zhang, H. Lyu, J. Tang, Z. Huang, Z. Du, X. Liu, A. Zeng, L. Hou, et al. LongBench: A bilingual, multitask benchmark for long context understanding. arXiv preprint arXiv:2308.14508, 2024.

S. R. Bowman and G. E. Dahl. What will it take to fix benchmarking in natural language understanding? In Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), pages 4843–4855, 2021.

D. T. Campbell and D. W. Fiske. Convergent and discriminant validation by the multitrait-multimethod matrix. Psychological Bulletin, 56(2):81–105, 1959.

D. Card, P. Henderson, U. Khandelwal, R. Jia, K. Mahowald, and D. Jurafsky. With little power comes great responsibility. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), pages 9263–9274, 2020.

S. Chaduvula, J. Ho, K. Kim, A. Narayanan, M. Alinoori, M. Garg, D. Ramachandram, and S. Raza. From features to actions: Explainability in traditional and agentic AI systems. arXiv preprint arXiv:2602.06841, 2026.

M. Chen, J. Tworek, H. Jun, Q. Yuan, H. P. de Oliveira Pinto, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman, A. Ray, R. Puri, G. Krueger, M. Petrov, H. Khlaaf, G. Sastry, P. Mishkin, B. Chan, S. Gray, N. Ryder, M. Pavlov, A. Power, L. Kaiser, M. Bavarian, C. Winter, P. Tillet, F. P. Such, D. Cummings, M. Plappert, F. Chantzis, E. Barnes, A. Herbert-Voss, W. H. Guss, A. Nichol, A. Paino, N. Tezak, J. Tang, I. Babuschkin, S. Balaji, S. Jain, W. Saunders, C. Hesse, A. N. Carr, J. Leike, J. Achiam, V. Misra, E. Morikawa, A. Radford, M. Knight, M. Brundage, M. Murati, K. Mayer, P. Welinder, B. McGrew, D. Amodei, S. McCandlish, I. Sutskever, and W. Zaremba. Evaluating large language models trained on code. arXiv preprint arXiv:2107.03374, 2021.

P. Clark, I. Cowhey, O. Etzioni, T. Khot, A. Sabharwal, C. Schoenick, and O. Tafjord. Think you have solved question answering? try ARC, the AI2 reasoning challenge. In arXiv preprint arXiv:1803.05457, 2018.

K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, and J. Schulman. Training verifiers to solve math word problems. arXiv preprint arXiv:2110.14168, 2021.

L. J. Cronbach and P. E. Meehl. Construct validity in psychological tests. Psychological Bulletin, 52(4):281–302, 1955.

A. D'Amour, K. Heller, D. Moldovan, B. Adlam, B. Alipanahi, A. Beutel, C. Chen, J. Deaton, J. Eisenstein, M. D. Hoffman, et al. Underspecification presents challenges for credibility in modern machine learning. Journal of Machine Learning Research, 23 (226):1–61, 2022.

K. de Langis, J. I. Park, B. Hu, K. C. Le, A. Schramm, M. C. Mensink, A. Elfenbein, and D. Kang. Strong memory, weak control: An empirical study of executive functioning in llms. arXiv preprint arXiv:2504.02789, 2025.

S. Fan, X. Ye, Y. Huo, Z.-Y. Chen, Y. Guo, S. Yang, W. Yang, S. Ye, J. Chen, et al. AgentProcessBench: Diagnosing step-level process quality in tool-using agents. arXiv preprint arXiv:2603.14465, 2026.

J. H. Flavell. Metacognition and cognitive monitoring: A new area of cognitive-developmental inquiry. American Psychologist, 34(10):906–911, 1979.

R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann. Shortcut learning in deep neural networks. Nature Machine Intelligence, 2 (11):665–673, 2020.

A. Ghasemabadi and D. Niu. Can llms predict their own failures? self-awareness via internal circuits. arXiv preprint arXiv:2512.20578, 2025.

C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In International Conference on Machine Learning (ICML), 2017.

F. G. Haznitrama, F. R. Ardi, and A. Oh. A neuropsychologically grounded evaluation of llm cognitive abilities, 2026. URL https://arxiv.org/abs/2603.02540.

D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt. Measuring massive multitask language understanding. In International Conference on Learning Representations (ICLR), 2021a.

D. Hendrycks, C. Burns, S. Kadavath, A. Arora, S. Basart, E. Tang, D. Song, and J. Steinhardt. Measuring mathematical problem solving with the MATH dataset. In Advances in Neural Information Processing Systems (NeurIPS), 2021b.

A. Z. Jacobs and H. Wallach. Measurement and fairness. In Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT), pages 375–385, 2021.

E. Jiang, C. Xu, N. Singh, T. Qiu, and G. Singh. Robust answers, fragile logic: Probing the decoupling hypothesis in llm reasoning, 2026. URL https://arxiv.org/abs/2505.17406.

M. K. Johnson, S. Hashtroudi, and D. S. Lindsay. Source monitoring. Psychological Bulletin, 114(1):3–28, 1993.

N. Kim and S. Schuster. Entity tracking in language models. arXiv preprint arXiv:2305.02363, 2023.

T. Lanham, A. Chen, A. Radhakrishnan, B. Steiner, C. Denison, D. Hernandez, D. Li, E. Durmus, E. Hubinger, J. Kernion, K. Lukosiute, K. Nguyen, N. Cheng, N. Joseph, N. Schiefer, O. Rauber, S. McCandlish, C. Olsson, T. Henighan, T. Hume, Z. Hatfield-Dodds, J. Kaplan, J. Clark, S. R. Bowman, and E. Perez. Measuring faithfulness in chain-of-thought reasoning. arXiv preprint arXiv:2307.13702, 2023.

B. Z. Li, Z. C. Guo, and J. Andreas. (how) do language models track state? arXiv preprint arXiv:2503.02854, 2025.

P. Liang, R. Bommasani, T. Lee, D. Tsipras, D. Soylu, M. Yasunaga, Y. Zhang, D. Narang, et al. Holistic evaluation of language models. Transactions on Machine Learning Research, 2023.

H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, and K. Cobbe. Let's verify step by step. arXiv preprint arXiv:2305.20050, 2023.

X. Liu, H. Yu, H. Zhang, Y. Xu, X. Lei, et al. AgentBench: Evaluating LLMs as agents. In International Conference on Learning Representations, 2023.

S. Messick. Validity. In R. L. Linn, editor, Educational Measurement, pages 13–103. American Council on Education / Macmillan, 3rd edition, 1989.

G. A. Miller. The magical number seven, plus or minus two: Some limits on our capacity for processing information. Psychological Review, 63(2):81–97, 1956.

A. Miyake, N. P. Friedman, M. J. Emerson, A. H. Witzki, A. Howerter, and T. D. Wager. The unity and diversity of executive functions and their contributions to complex "frontal lobe" tasks. Cognitive Psychology, 41(1):49–100, 2000.

T. O. Nelson and L. Narens. Metamemory: A theoretical framework and new findings. Psychology of Learning and Motivation, 26:125–173, 1990.

M. Nye, A. J. Andreassen, G. Gur-Ari, H. Michalewski, J. Austin, D. Biber, D. Dohan, A. Lewkowycz, M. Bosma, D. Luan, C. Sutton, and A. Odena. Show your work: Scratchpads for intermediate computation with language models. arXiv preprint arXiv:2112.00114, 2021.

H. Putnam. Psychological predicates. In W. H. Capitan and D. D. Merrill, editors, Art, Mind, and Religion, pages 37–48. University of Pittsburgh Press, 1967.

K. Rezaee, J. Camacho-Collados, and M. T. Pilehvar. Exploring state tracking capabilities of large language models. arXiv preprint arXiv:2511.10457, 2025.

M. T. Ribeiro, T. Wu, C. Guestrin, and S. Singh. Beyond accuracy: Behavioral testing of NLP models with CheckList. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL), pages 4902–4912, 2020.

N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao. Reflexion: Language agents with verbal reinforcement learning. In Advances in Neural Information Processing Systems, volume 36, 2023.

R. H. Somers. A new asymmetric measure of association for ordinal variables. American Sociological Review, 27(6):799–811, 1962.

J. Sweller. Cognitive load during problem solving: Effects on learning. Cognitive Science, 12(2):257–285, 1988.

E. Tulving et al. Episodic and semantic memory. Organization of memory, 1(381-403):1, 1972.

M. Turpin, J. Michael, E. Perez, and S. R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. arXiv preprint arXiv:2305.04388, 2023.

J. Uesato, N. Kushman, R. Kumar, F. Song, N. Siegel, L. Wang, A. Creswell, G. Irving, and I. Higgins. Solving math word problems with process- and outcome-based feedback. arXiv preprint arXiv:2211.14275, 2022.

C. Wang and J. V. Sun. Unable to forget: Proactive interference reveals working memory limits in llms beyond context length, 2025. URL https://arxiv.org/abs/2506.08184.

W. Wang, J. Min, and W. Zou. Intelligence degradation in long-context llms: Critical threshold determination via natural length distribution analysis. arXiv preprint arXiv:2601.15300, 2026.

W. Xia et al. Minerva: A programmable memory test benchmark for language models. In Proceedings of the 42nd International Conference on Machine Learning, volume 267 of Proceedings of Machine Learning Research, 2025. URL https://proceedings.mlr.press/v267/xia25c.html.

S. Xie, Z. Shi, H. Shen, G. Huang, Y. Ma, and X. Jing. M3-bench: Process-aware evaluation of llm agents social behaviors in mixed-motive games. arXiv preprint arXiv:2601.08462, 2026.

Z. Xu, Z. Wu, Y. Zhou, A. Feng, et al. Beyond correctness: Rewarding faithful reasoning in retrieval-augmented generation. arXiv preprint arXiv:2510.13272, 2025.

Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. W. Cohen, R. Salakhutdinov, and C. D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In Proceedings of EMNLP 2018, 2018.

S. Yao, H. Chen, J. Yang, and K. Narasimhan. WebShop: Towards scalable real-world web interaction with grounded language agents. In Advances in Neural Information Processing Systems (NeurIPS), 2022.

R. Zellers, A. Holtzman, Y. Bisk, A. Farhadi, and Y. Choi. HellaSwag: Can a machine really

finish your sentence? In Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL), 2019.

S. Zhang et al. Working memory identifies reasoning limits in language models. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pages 13421–13434, 2024. URL https://aclanthology.org/2024.emnlp-main.938/.

A Full Related Work Table

Table 10. Representative benchmarks and evaluation work, grouped by measurement type and CEF gap addressed.

Work	Measures	CEF gap addressed
MMLU [Hendrycks et al., 2021a], ARC [Clark et al., 2018], HellaSwag [Zellers et al., 2019]	Knowledge recall	No memory structure, no process
HumanEval [Chen et al., 2021], MATH [Hendrycks et al., 2021b], GSM8K [Cobbe et al., 2021]	Output correctness	No reasoning fidelity
HotpotQA [Yang et al., 2018], WebShop [Yao et al., 2022]	Task completion	No metacognition, no memory typ
AgentBench [Liu et al., 2023]	Sequential task success	No within-task process
Reflexion [Shinn et al., 2023]	Post-reflection improvement	Monitoring $\neq$ control (conflated)
PAE [Xie et al., 2026]	Corruption rate	Motivates CEF; no per-dimension measurement
NeuroCognition [Hazmitrama et al., 2026]	Cognitive ability (IQ)	Ability $\neq$ process quality; no load variation
Strong Mem. & Weak Ctrl. [de Langis et al., 2025]	Executive function	Isolated tasks; no multi-turn load
Entity Tracking [Kim and Schuster, 2023]	Entity-state	Prior art for WMF-AM design (passive retrieval)
Latent State Persistence [Anonymous, 2025a]	Sequential latent-state manipulation	Closest prior paradigm; no arithmetic load, no agent battery
Minerva [Xia et al., 2025]	Programmable memory (incl. Quantity State: sequential +/-)	Closest arithmetic prior; WMF-AM adds construct-isolation suite + agent-predictive validity
Zhang et al. [Zhang et al., 2024]	WM-style probes $\rightarrow$ reasoning limits	Motivational; no ablation suite, no agent battery
Intell. Degradation [Wang et al., 2026]	Load degradation	Prior art for CLA
HELM [Liang et al., 2023]	Behavioral outcomes	No cognitive processes
Long-context [Bai et al., 2024]	Retrieval from static window	WMF-AM targets active transformation
ECE calibration [Guo et al., 2017]	Confidence scalar	MCC decomposes into monitoring/control

B Development Status and Summary Tables

Table 11. Development status of all 11 CEF sub-dimensions. Status key—A: robust differentiation + multi-seed reliability; B: differentiation shown, limited seeds/models; C: probe redesigned, initial results; D: ceiling/floor effect, needs redesign; E: protocol specified, no pilot data.

Sub-dim	Status	Evidence	Limitation
WMF-AM	A^+	Δ=0.933, 4-seed all 20, 3-template (τ=0.524), yoked control, predictive τ=0.612	All open-weight
WMF-IM	B	spread 0.480, N=5	Single paradigm (hard)
WMF-IR	D	Ceiling (= 1.0 all models)	Needs redesign
MCC-MA	B	4/7 models differentiated	Floor for 3 models
MCC-CE	C	v2: error 23–40%, near-zero flagging (N=8)	Binary protocol
EMC-lite	B	Source-conflict spread 0.400	Single seed
CLA-DC	$B^\dagger$	Correlates τ=0.905 with WMF-AM	†Provisionally collapsed onto WMF
CLA-CR	C	v3 spread 1.171, N=7	v1–v2 failed; 2 seeds
CLA-RA	E	—	Awaits full study
EMC-TO	E	—	Awaits full study
EMC-EI	E	—	Awaits full study

Table 12. CEF at a glance. Each sub-dimension's task type, scoring rule, and pilot status. Tier 1 = novel construct; Tier 2 = novel reframing. Pilot: ✓ = pilot-supported, $\sim$= preliminary, —= protocol only.

Sub-dim	Task (one-line)	Scoring	Tier	Pilot
WMF-AM	Track K sequential state updates, report final state	Exact match	1	✓
WMF-IM	Recall N facts amid M distractors	Recall accuracy	2	$\sim$
WMF-IR	Recall List-A after List-B interference	Proactive interference δ	2	—
MCC-MA	Predict which of 20 answers are wrong	Pearson r (predicted vs. actual)	1	$\sim$
MCC-CE	Flag uncertain answers, revise them	Correction efficacy	1	$\sim$
EMC-TO	Reorder non-chronological events	Kendall τ vs. ground truth	1	—
EMC-EI	Discriminate similar episodes	Accuracy under similarity	1	—
EMC-SA	Attribute facts to source + episode	Joint attribution accuracy	2	—
CLA-DC	Performance across 5 load levels	Smoothness (SD of Δs)	1	—
CLA-RA	Response elaboration vs. difficulty	Pearson r	1	—
CLA-CR	Easy-task recovery after high-load block	Recovery rate	1	$\sim$

C Glossary

D On Cognitive-Science Labels

We use labels like "working memory" and "metacognition" as operational shorthand for specific behavioral probes, not as claims about LLM internal mechanisms. "WMF-AM"

Table 13. Glossary of key terms and acronyms.

Term	Definition
CEF	Cognitive Evaluation Framework (this paper)
WMF-AM/IM/IR	Working Memory Fidelity: Active Manipulation / Information Maintenance / Interference Resistance
MCC-MA/CE	Metacognitive Calibration: Monitoring Accuracy / Control Efficacy
EMC-TO/EI/SA	Episodic Memory Coherence: Temporal Ordering / Episodic Interference / Source Attribution
CLA-DC/RA/CR	Cognitive Load Adaptation: Degradation Curve / Resource Allocation / Capacity Recovery
AGENT-PQ/TC	Agent Process Quality / Task Completion (external evaluation)
CONV-RFPOC	Convergent validity probe: Reasoning Fidelity–Product of Cognition

means: "accuracy on sequential state-tracking tasks degrades with depth K"; it does not mean LLMs have a working memory buffer. "MCC-MA" means: "accuracy of self-error prediction on a 20-item probe"; it does not mean LLMs have phenomenal metacognition. The labels are justified because the probes are adapted from validated human measurement paradigms, and they organize what would otherwise be an unstructured collection of tests.

Two caveats are essential. First, we do not require LLMs to have phenomenal cognition or internal states analogous to human working memory. CEF dimensions are process-sensitive behavioral probes that borrow measurement logic from cognitive science. Second, CEF measurements are behavioral (prompt/response), designed to be differentially sensitive to process quality: high MCC-MA requires accurate self-error prediction; graceful CLA-DC requires proportional rather than catastrophic degradation.

E Full Sub-dimension Definitions

WMF-AM (Active Manipulation, Tier 1). Uses a complex span task: initial state, K sequential modifications, report final state. Depths $K \in \{2, 4, 6, 8, 12\}$, three surface forms. Prior entity tracking [Kim and Schuster, 2023] tests passive retrieval; WMF-AM tests sequential transformation.

WMF-IM (Information Maintenance, Tier 2). Embeds N facts in a passage with M distractors; measures recall accuracy. Extends Wang and Sun [Wang and Sun, 2025] to natural-language multi-turn contexts.

WMF-IR (Interference Resistance, Tier 2). Adapts the List-A/List-B proactive interference paradigm. Currently shows ceiling (= 1.0 all models); needs redesign.

MCC-MA (Monitoring Accuracy, Tier 1). Model predicts which of 20 answers are incorrect; Pearson r measures monitoring accuracy. Unlike Gnosis [Ghasemabadi and Niu, 2025] (requires internals), MCC-MA uses prompting under multi-turn load.

MCC-CE (Control Efficacy, Tier 1). Model flags uncertain answers and revises them. Decomposes into flagging rate, correction efficacy, and false alarm rate.

EMC-TO/EI/SA. Temporal ordering (Kendall τ), cross-episode discrimination under similarity, and joint source-episode attribution. All proposed, no pilot data.

CLA-DC (Degradation Curve, Tier 1). Smoothness score (SD of successive Δs). Provisionally collapsed onto WMF-AM ($\tau = 0.905$).

CLA-RA/CR. Resource allocation (difficulty–elaboration correlation) and capacity recovery (post-load accuracy recovery). CLA-RA has no pilot data; CLA-CR-v3 spread = 1.171.

Composites (all exploratory). WMF = $0.50\times$AM$+0.30\times$IM$+0.20\times$IR; MCC = $0.55\times$MA$+0.45\times$CE; EMC = $0.40\times$EI$+0.35\times$TO$+0.25\times$SA; CLA = $0.50\times$DC$+0.30\times$RA$+0.20\times$CR. We do not propose an overall CEF composite; per-dimension profiles are the primary unit of analysis.

Table 14. Construct map for validated probes: manipulated factor, observable signature, rival explanations, and falsification criteria.

Probe	Manipulated factor	Observable signature	Plausible rivals	Expected pattern	Falsified if
WMF-AM	Depth K (number of sequential state updates)	Accuracy × depth interaction	Instruction following; tokenizer parsing; template familiarity; long-context retrieval	Low τ with completion; monotonic depth decline; $\tau_{\text{bare,chat}} > 0.4$; yoked control $\gg$ WMF-AM	High τ with completion ($\geq$ 0.55); no depth effect; template-specific rankings; control degrades at same rate
MCC-MA	Error-laden vs correct items	Pearson r (predicted vs actual errors)	Calibration shortcuts; always-predict-correct bias	Low τ with WMF-AM; model-specific prediction	Uniform predictions; $\tau \approx 1$ with WMF-AM
EMC-lite	Conflict vs control episodes	Source attribution accuracy under interference	Recency bias; keyword matching	Low τ with WMF-AM; fault > control difficulty	No fault–control difference; $\tau \approx 1$ with WMF-AM

F Construct-to-Probe Mapping and Falsifiability

G Rival Explanations

For each probe, plausible alternative explanations remain open. WMF-AM: (a) Instruction following—models may fail at depth because they lose track of instructions. The depth gradient partly addresses this. (b) Long-context retrieval—the control addresses this for 5/7 models. (c) Template artifacts—cross-template stability (τ = 0.685–0.781) mitigates but does not eliminate this. MCC-MA/CE: (a) Calibration shortcuts. (b) Prompt sensitivity. EMC-lite: Recency bias; keyword matching. These are the primary targets for the $N \geq 15$ study.

H Study Matrix

Table 15. Study matrix: all experiments, models, and sample sizes. †llama3.1:70b added for MCC-CE-v2 only.

Experiment	Metric	N_{models}	Seeds	Trials/model
Outcome Correctness (v2)	Accuracy	20	1	100
WMF-AM (dissociation)	Exact match	20	4	180
Agent Validation	Process quality	20	1	10
Template Harmonization	Rank stability	15	1	135
Yoked Cancellation Control	Exact match	15	1	100
MCC-MA	Pearson r	15	1	20
MCC-CE-v2	Flagging rate	8†	1	50
EMC-lite	Composite	7	1	18
CLA-CR-v3	Recovery rate	7	2	7 blocks
AGENT-PQ/TC	Rubric / binary	7	1	10
CONV-RFPOC	Fidelity	7	1	15

I Yoked Cancellation Control Details

The yoked cancellation control uses operation pairs that cancel: "Alice gains 3 points" followed immediately by "Alice loses 3 points." The prompt format, entity names, and depth structure match WMF-AM exactly, so the arithmetic parsing demand is identical. The correct answer equals the initial state, but unlike the original inert control, the model must process each operation to verify cancellation. This eliminates the trivial shortcut of ignoring operations.

J Measurement Robustness

Multi-seed reliability. All 15 models evaluated with 4 independent random seeds × 15 probes. Original-model cross-seed $\tau = 0.798$. Expansion-model SDs: phi3:14b = 0.094, gemma2:9b = 0.144, qwen2.5:3b = 0.122, llama3.2:3b = 0.122, deepseek-r1:7b = 0.114, mixtral:8x7b = 0.116, command-r:35b = 0.100, yi:34b = 0.084 (mean SD = 0.108).

Cross-template stability. Phase 5 administered WMF-AM under three prompt wrappers (bare, chat, chain-of-thought) across all 15 models. Bare ↔ chat: $\tau = 0.524$ ($p = 0.006$)—significant rank preservation. Bare ↔ CoT: $\tau = 0.105$ ($p = 0.627$)—CoT pushes all models to ceiling (≥ 0.93), eliminating variance. Chat ↔ CoT: $\tau = 0.314$ ($p = 0.114$). The CoT ceiling effect is interpretively meaningful: externalized reasoning via scratchpad bypasses the working-memory bottleneck that WMF-AM is designed to measure.

Non-ceiling completion proxies. τ(WMF-AM, MMLU) = 0.356 ($p = 0.317$; range 0.25–0.72); τ(WMF-AM, GSM8K) = 0.150 ($p = 0.645$; range 0.05–0.83). Scores from published model cards, not re-evaluated under pilot setup.

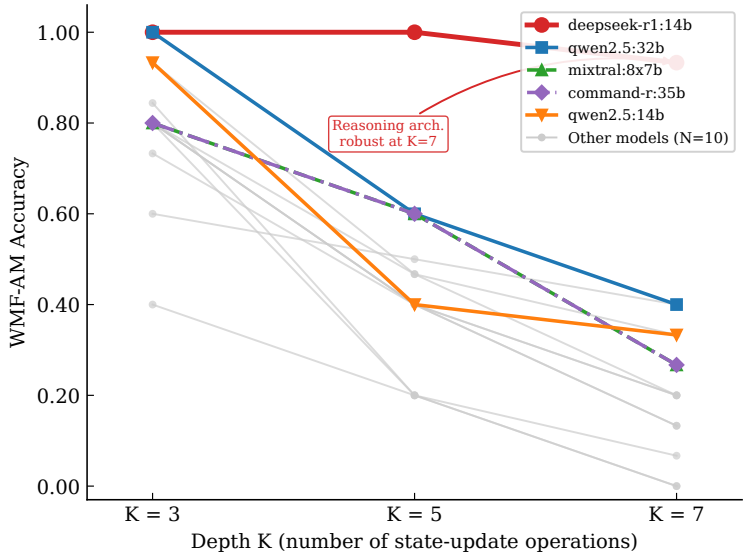

Figure 4. WMF-AM accuracy by depth K (multi-seed means, N=7). deepseek-r1:14b maintains near-perfect accuracy; others degrade at K=7.

Depth profile.

K EMC-lite Results

Model	F-Det	F-Src	F-Rec	F-Comp	C-Rec	ΔRec
qwen2.5:14b	0.917	0.750	0.917	0.917	0.972	−0.056
qwen2.5:32b	0.917	0.750	0.917	0.917	1.000	−0.083
deepseek-r1:14b	0.889	0.917	0.917	0.912	1.000	−0.083
llama3.1:8b	0.778	0.583	0.750	0.771	0.889	−0.139
qwen2.5:7b	0.861	0.583	0.639	0.736	0.722	−0.083
mistral:7b	0.750	0.417	0.639	0.685	0.722	−0.083
gemma2:27b	0.861	0.583	0.528	0.667	0.611	−0.083
Δ	0.167	0.500	0.389	0.250	0.389	—

Table 16. EMC-lite (N=7, hardened protocol). Source attribution shows strongest differentiation (spread = 0.500).

EMC-lite provides preliminary support for the source-conflict slice. It does not establish EMC as a whole; single-seed evidence only. τ(WMF-AM, EMC-lite) = 0.250: substantial independence.

L MCC-CE-v2 Results

Table 17. MCC-CE-v2 (N=8). Error rates 23–40% (floor eliminated); flagging near zero for all models.

	Balanced (n=30)		Trap-only (n=20)	
Model	Error%	Flag%	Error%	Flag%
qwen2.5:7b	26.7	0.0	40.0	0.0
qwen2.5:14b	36.7	0.0	40.0	0.0
qwen2.5:32b	30.0	0.0	25.0	0.0
llama3.1:8b	26.7	0.0	40.0	0.0
llama3.1:70b	—	—	25.0	0.0
gemma2:27b	30.0	11.1	30.0	0.0
deepseek-r1:14b	23.3	0.0	35.0	0.0
mistral:7b	30.0	0.0	40.0	0.0

The v1 floor was an artifact of easy questions; v2 models make many errors but show near-zero self-flagging under the binary protocol. Alternative designs (continuous confidence, forced ranking) might reveal partial capacity.

M Negative OOD Result

Two OOD stress-test designs (v1: 5-step NL tasks; v2: K=8–9 interference tasks) produced anti-correlated CEF–OOD relationships across model pairs. We make no predictive claim. Full assessment requires $N \geq 11$ with process-sensitive downstream tasks.

N Full Validity Analyses

The validity summary table appears in the main text (Table 8). Below we provide additional details.

Convergent validity details. AGENT-PQ: 10 multi-step scenarios scored by LLM judge (GPT-4o) on 4 rubric dimensions. CONV-RFPOC: 15 reasoning problems with counterfactual chain perturbation. RFPOC = $1 - P$(same answer | perturbed chain). Full protocol in Appendix O. Limitation: LLM judge may introduce shared bias.

Construct separation (WMF-AM $\leftrightarrow$ CLA-DC). $\tau = 0.905$: no evidence of separability at N=7. Both involve state tracking under load. Planned discriminant test at $N \geq 15$. CLA-CR-v3 provides a structurally independent CLA measure. Even if CLA-DC merges with WMF-AM, WMF-AM, MCC-MA, and EMC-lite remain separable ($\tau \leq 0.250$).

Sensitivity. LOMO for CONV-RFPOC vs AGENT-PQ: τ ranges 0.600–1.000 (mean 0.833). Leave-one-family-out (Qwen, N=4): $\tau = 0.667$. After Holm correction, RFPOC–AGENT-PQ remains significant ($p_{\text{adj}} = 0.021$).

Multiple testing. Three primary hypotheses; all others exploratory.

Benchmark governance. All headline results use frozen protocol versions. Probe revisions were motivated by floor/ceiling effects, not model-specific tuning.

O AGENT-PQ Evaluation Protocol

AGENT-PQ uses 10 multi-step scenarios where each model operates within a tool-use scaffold. A separate LLM judge (GPT-4o, temperature = 0) scores each trajectory on four rubric dimensions: plan coherence, state tracking, error recovery, goal adherence. AGENT-PQ = mean of 4 rubric scores/4. AGENT-TC is the binary success rate on the same scenarios.

CONV-RFPOC administers 15 reasoning problems with counterfactual chain perturbation. RFPOC = $1 - P$(same answer | perturbed chain).

P Complete Experimental Protocols

Full prompt templates, parameter specifications, scoring procedures, and reference implementations for all 11 sub-dimensions are provided in the supplementary materials.

Q Weight Sensitivity Analysis

Three weighting schemes (proposed, equal, PCA-derived) are evaluated. Per-dimension profiles are primary; composites are exploratory only.

R Dissociation Case Studies

Matched dissociation pairs: deepseek-r1:14b and qwen2.5:14b share similar EMC-lite (0.912 vs. 0.917) but differ on WMF-AM (0.983 vs. 0.467). mistral:7b and llama3.1:8b have similar WMF-AM but differ on EMC-lite. These cross-dimension pairs anchor the multiple realizability argument.

S Worked Example: WMF-AM at Depth K=3

Setup. "A warehouse has 3 items: Widget=50, Gadget=30, Sprocket=20." Three updates: "Ship 10 Widgets," "Receive 25 Gadgets," "Transfer 5 Sprockets to overflow."
Correct. Widget=40, Gadget=55, Sprocket=15.
Scoring. Exact match per item. deepseek-r1:14b scores 1.000 at K=7; llama3.1:8b scores 0.050. Both score 0.90–0.95 on outcome correctness. This is the Completion Fallacy in action.

T Novelty Classification

Table 18. Novelty tier for all 11 sub-dimensions. Tier 1 = novel; Tier 2 = novel reframing.

Sub-dim	Tier	Prior art / differentiation
WMF-AM	1	Entity Tracking [Kim and Schuster, 2023]: passive retrieval; Latent State Persistence [Anonymous, 2025a]: structural ops, no arithmetic load, no agent battery; WMF-AM = cumulative arithmetic transformation + ablation suite + downstream validity
WMF-IM	2	Wang & Sun [Wang and Sun, 2025]: key-value pairs; WMF-IM = NL multi-turn
WMF-IR	2	Extended to domain-varied materials
MCC-MA	1	Gnosis [Ghasemabadi and Niu, 2025] requires internals; MCC-MA = prompting only
MCC-CE	1	Reflexion [Shinn et al., 2023]: aggregate; MCC-CE decomposes detection/correction
EMC-TO	1	No prior LLM temporal ordering measurement
EMC-EI	1	No prior cross-episode interference diagnostic
EMC-SA	2	Interference-conditioned attribution
CLA-DC	1	Intell. Degradation [Wang et al., 2026]: magnitude; CLA-DC = shape
CLA-RA	1	Not previously measured
CLA-CR	1	Not previously measured